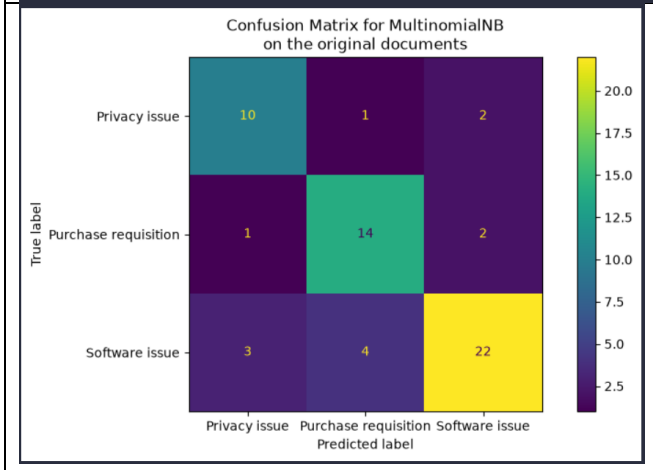
```python
from sklearn.feature_extraction.text import CountVectorizer
vectorizer = CountVectorizer(analyzer = 'word',
                             max_features = 5000,
                             ngram_range=(1,2))
train_data_features = vectorizer.fit_transform(clean_train_tickets)
#Create a CountVectorizer object
```

✓ 0.0s

```
0.7796610169491526
                    precision    recall  f1-score   support

     Privacy issue       0.71      0.77      0.74        13
Purchase requisition     0.74      0.82      0.78        17
      Software issue     0.85      0.76      0.80        29

          accuracy                           0.78        59
         macro avg       0.77      0.78      0.77        59
      weighted avg       0.79      0.78      0.78        59
```



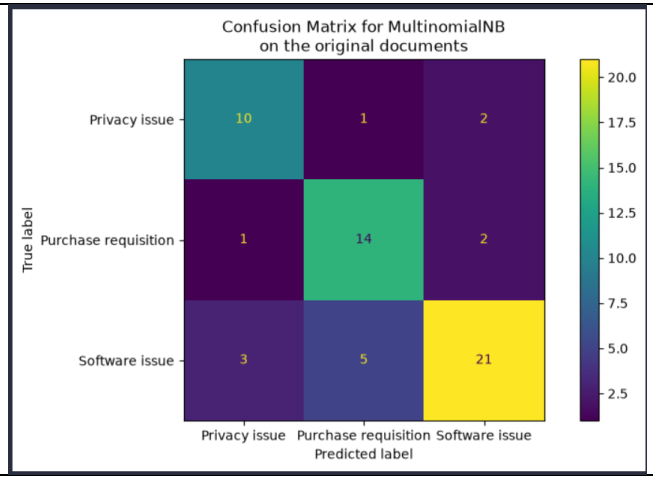

```python
from sklearn.feature_extraction.text import CountVectorizer
vectorizer = CountVectorizer(analyzer = 'word',
                             max_features = 1000,
                             ngram_range=(1,1))
train_data_features = vectorizer.fit_transform(clean_train_tickets)
```

✓ 1.3s

```
0.7627118644067796
                    precision    recall  f1-score   support

     Privacy issue       0.71      0.77      0.74        13
Purchase requisition     0.70      0.82      0.76        17
      Software issue     0.84      0.72      0.78        29

          accuracy                           0.76        59
         macro avg       0.75      0.77      0.76        59
      weighted avg       0.77      0.76      0.76        59
```




```python
from sklearn.feature_extraction.text import CountVectorizer
vectorizer = CountVectorizer(analyzer = 'word',
                             max_features = 1000,
                             ngram_range=(1,2))
train_data_features = vectorizer.fit_transform(clean_train_tickets)
```

✓ 1.4s

```
0.7457627118644068
                    precision    recall  f1-score   support

     Privacy issue       0.69      0.69      0.69        13
Purchase requisition     0.70      0.82      0.76        17
      Software issue     0.81      0.72      0.76        29

          accuracy                           0.75        59
         macro avg       0.73      0.75      0.74        59
      weighted avg       0.75      0.75      0.75        59
```

